

Shared objective: $U_t(b) = b\, g^{+}(1-\rho_t) \;-\; (1-b)\, g^{-} \;-\; \lambda_l\, e_t \;-\; \lambda_c\,(1+w_t), \quad U(\text{skip}) = 0$

(a) Joint (Ours)

at each partial k, with b_k and $\hat{R}_k$:
dispatch now if $U_k > 0$ and $U_k \geq \text{Wait}_k$,
else wait for the next partial;
at end of turn: retrieve iff $U_K > 0$, else skip

whether and when are chosen together: the dispatch time depends on the belief

(b) Decomposed (B9, B9+)

timing rule
fixed trigger τ,
set before the turn
(no belief)

retrieval rule
at τ: retrieve
iff $U(b) > 0$

no shared state; B9 prices the full latency L, B9+ the expected exposure of τ

(c) Speculative (B10, B11)

prefetch at first
query-ready partial
(B10 always;
B11 if belief gate)

end of turn:
use result
iff $U(b_K) > 0$

an unused prefetch costs compute only; (a) also runs under this architecture